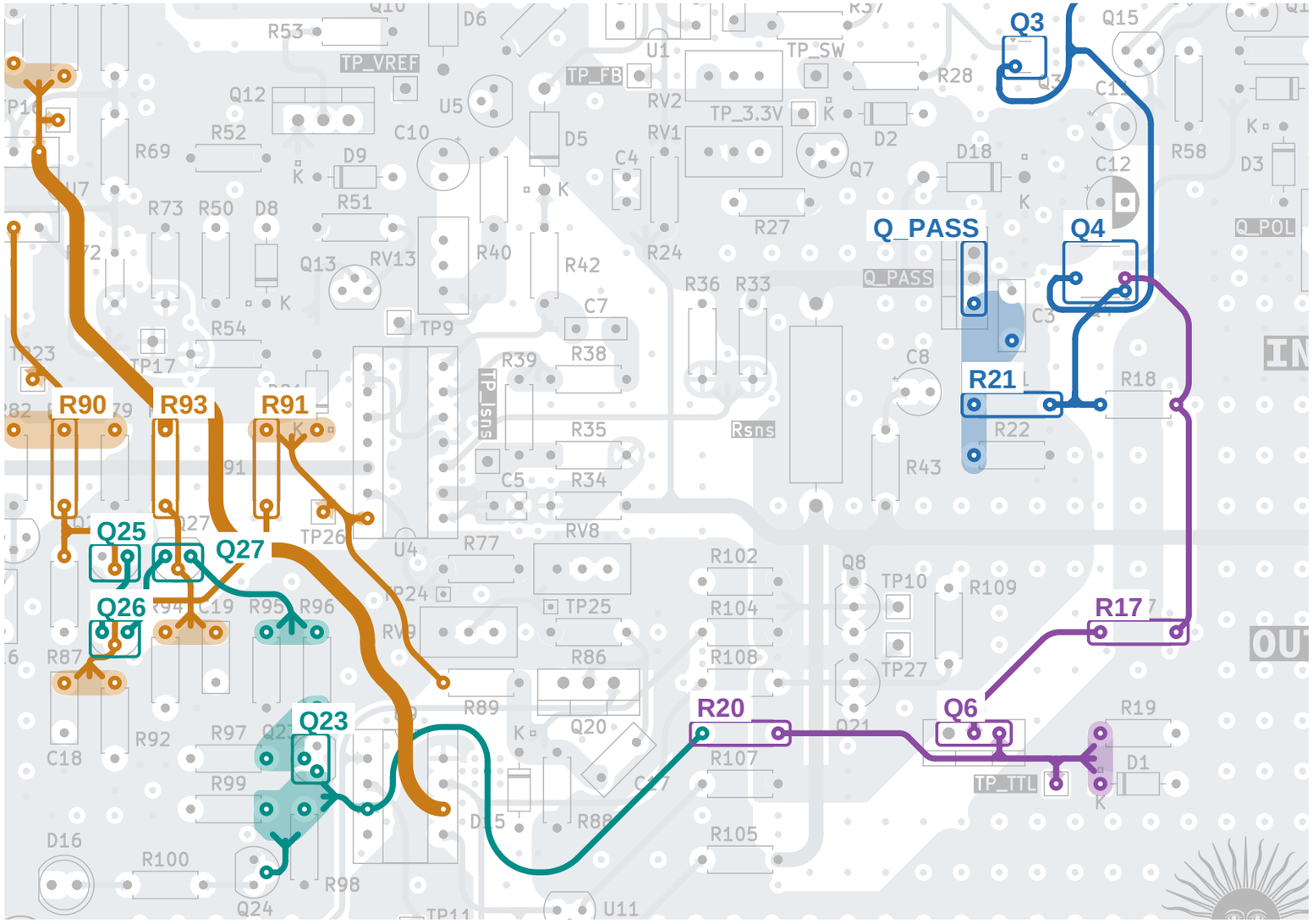
Enable logic and pass-drive switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



Permission inputs

PCB nets 1 (UVLO), 2 (OVLO), and LDO2 (delay) drive the series NPN chain.

Discrete AND function

Q25-Q27 and the inverter Q23 generate the TTL enable node.

Gate control

TTL drives Q6 through R20. Q6 controls Q4A through R17.

Pass-base drive

Q4A connects the Q3 output-stage path through R21 to the pass base.

Q4A and Q4B share the Q4 footprint. This view highlights the enable section Q4A.

UVLO / OVLO / delay → Q25-Q27 → Q23: TTL → R20 / Q6 → R17 / Q4A → R21 / pass base

Control sequence only. The load-current path passes through Q4B and Q_PASS, not through this logic chain.

Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.